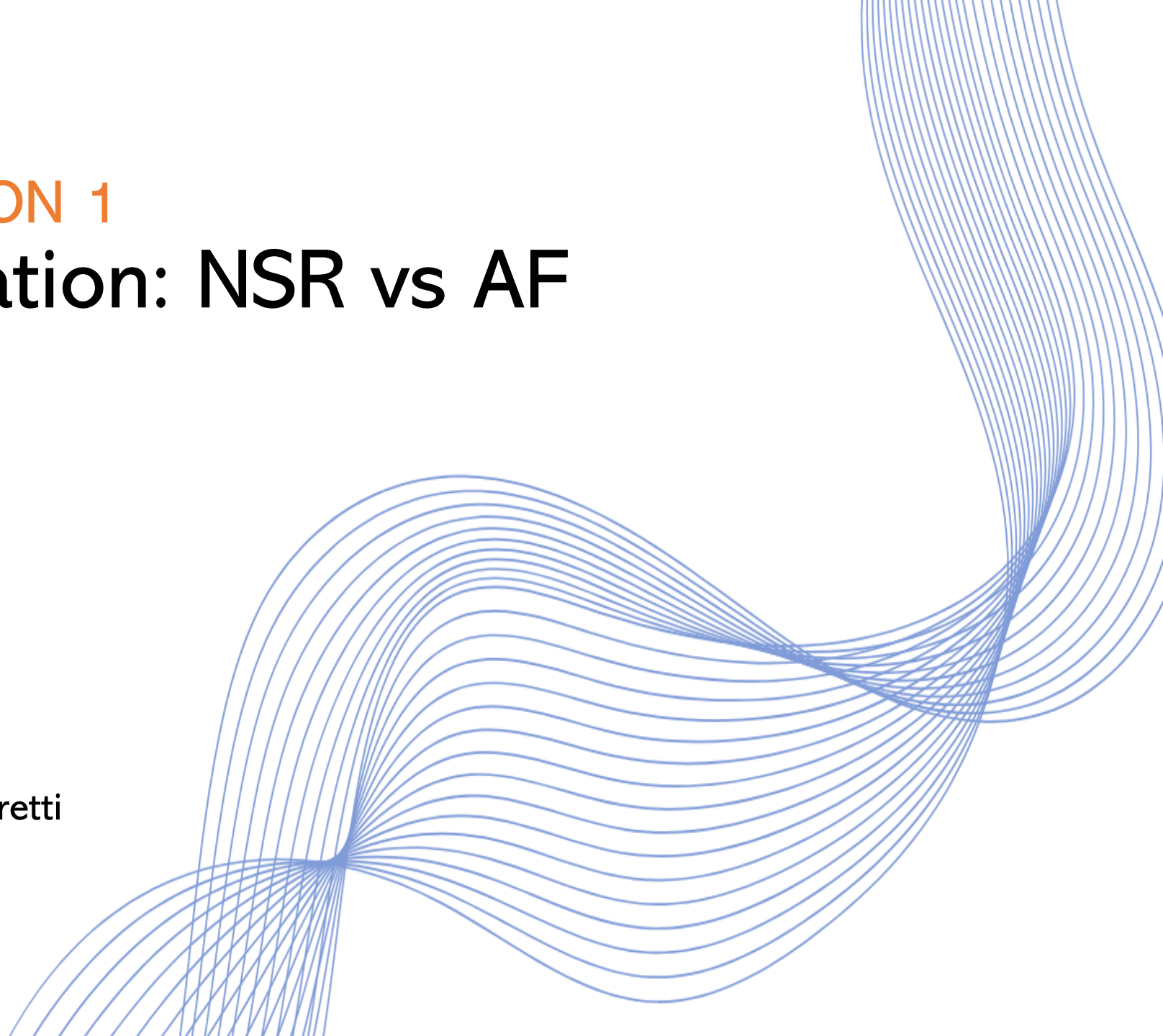


PRACTICAL SESSION 1

ECG classification: NSR vs AF

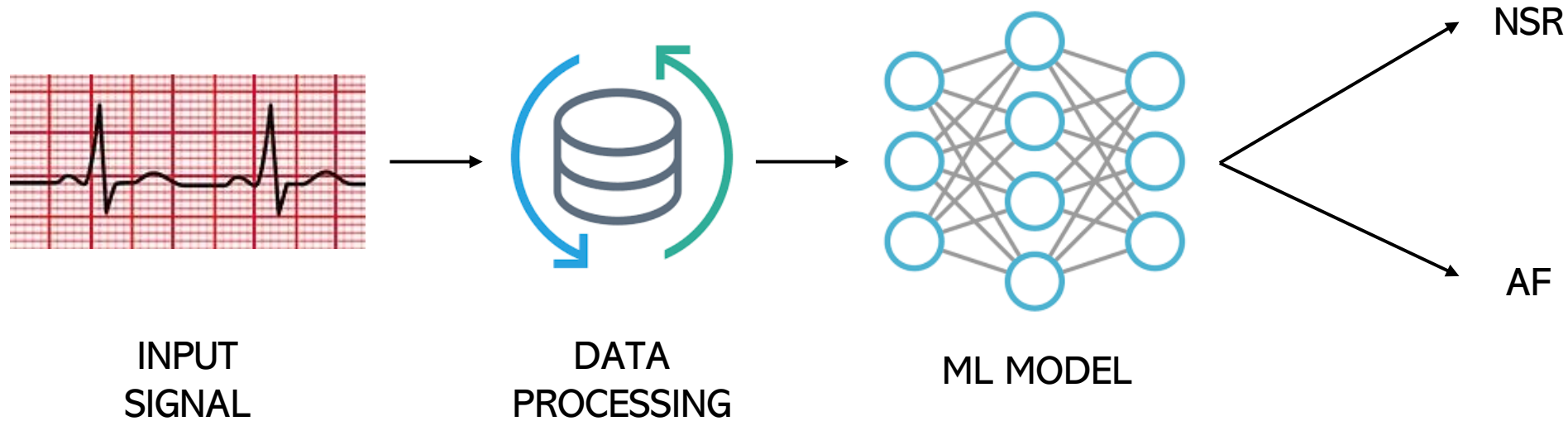
Professor – Valentina Corino
valentina.corino@polimi.it

Teaching assistant – Meri Ferretti
meri.ferretti@polimi.it



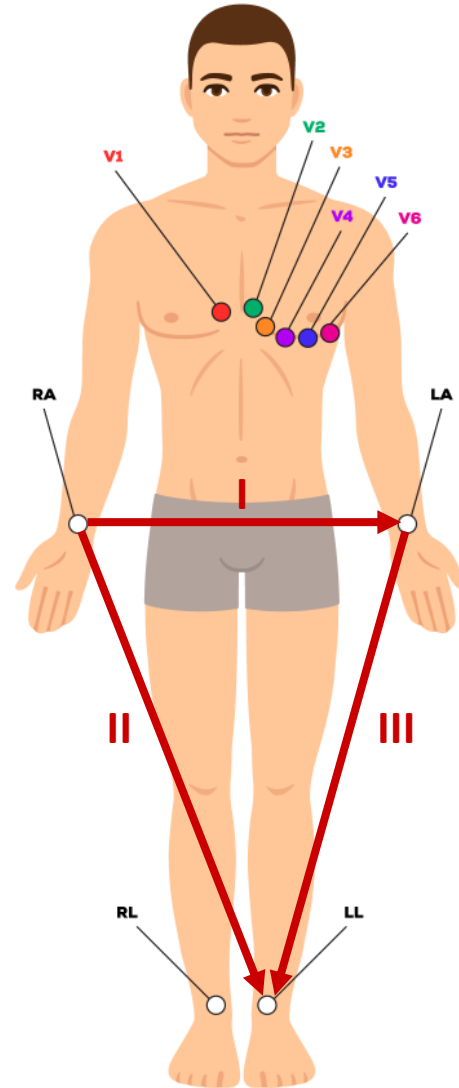
Objective of the day

→ To develop a ML model able to distinguish NSR from AF

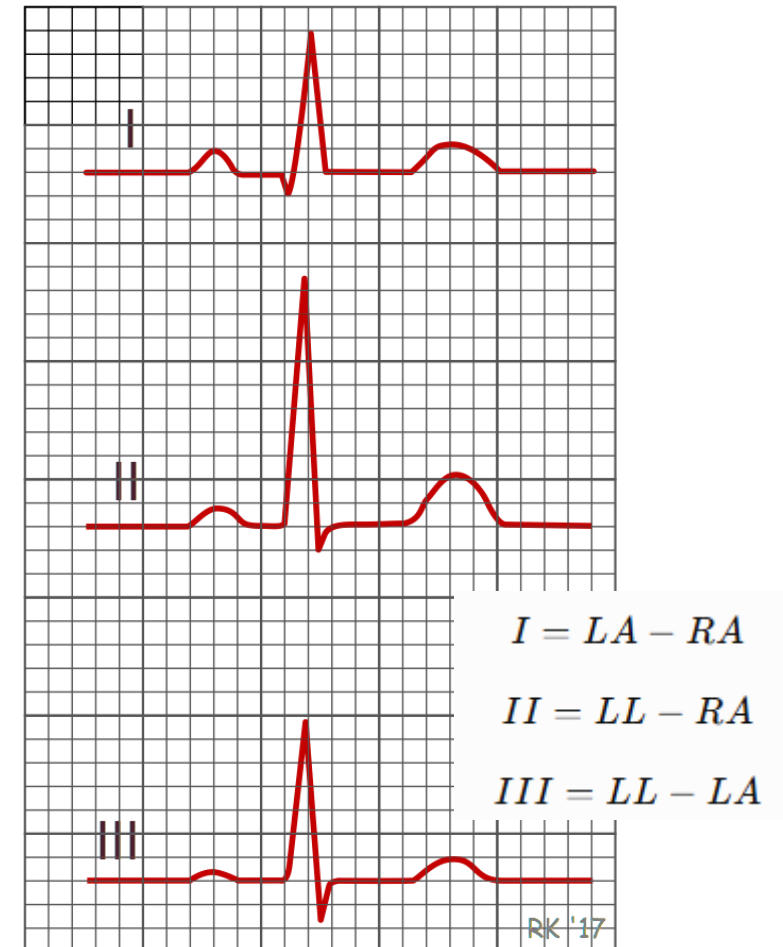


ECG recording

- ECG: recording of the heart's electrical activity
- It is recorded through electrodes placed in the torso and limbs
- Electrodes measures the electrical activity of the heart from different orientations (leads)

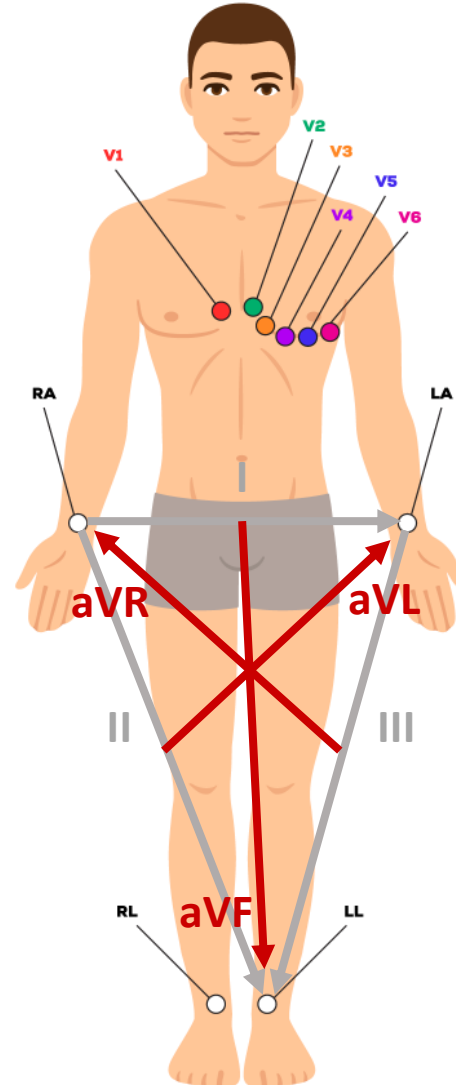


Limb leads (bipolar)

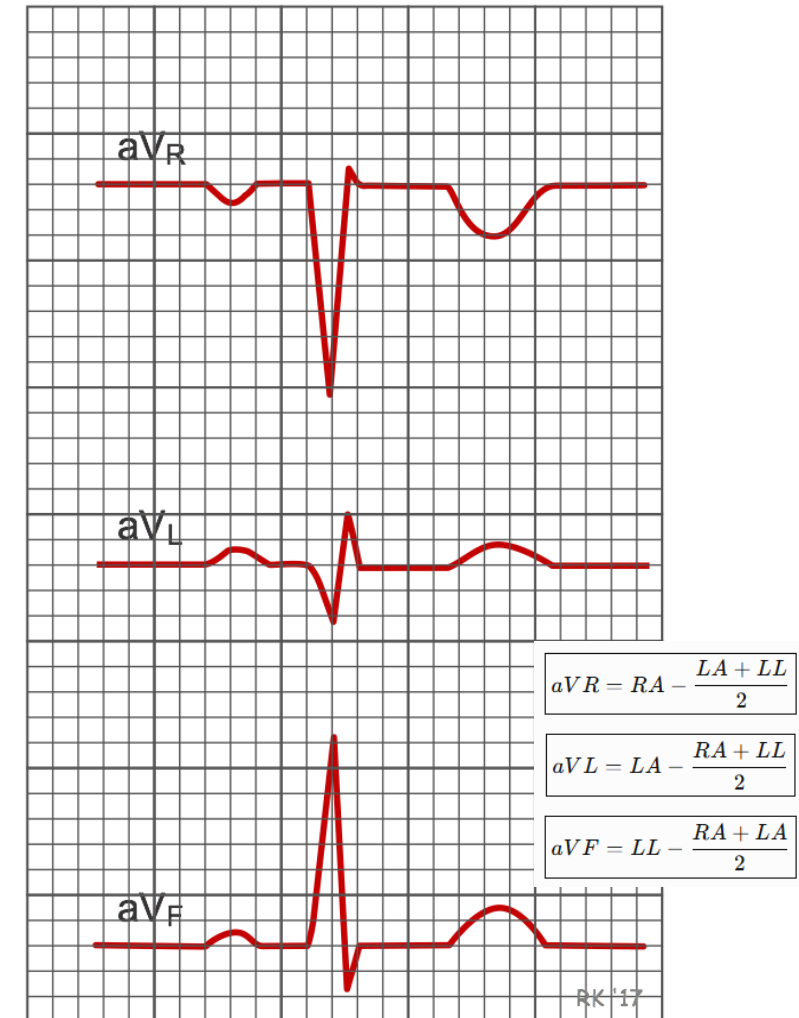


ECG recording

- ECG: recording of the heart's electrical activity
- It is recorded through electrodes placed in the torso and limbs
- Electrodes measures the electrical activity of the heart from different orientations (leads)

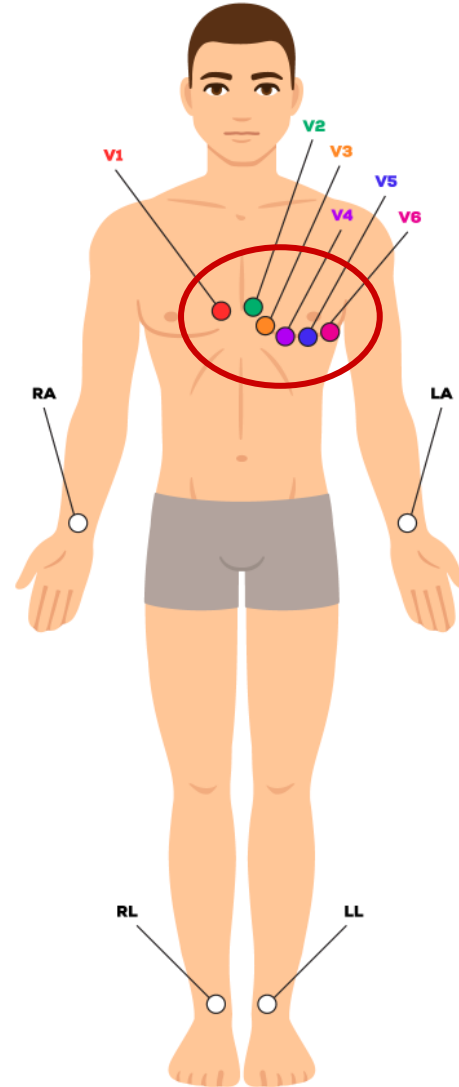


Augmented Limb leads (unipolar)

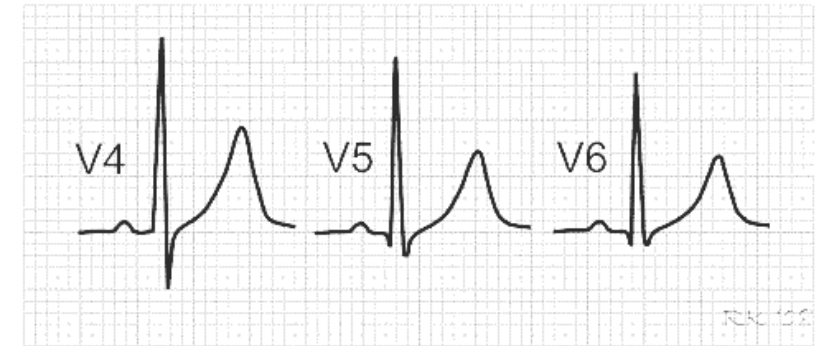
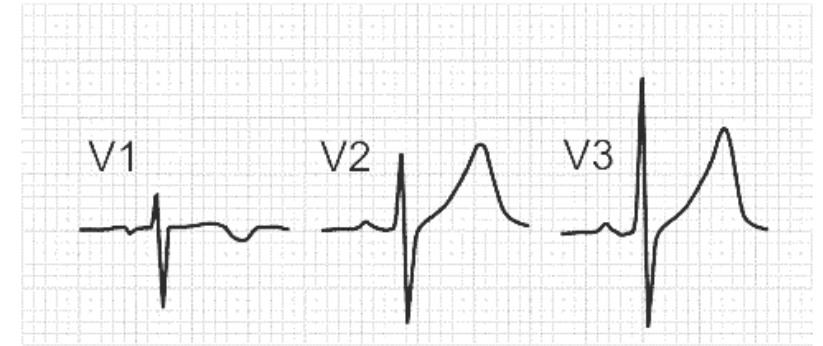


ECG recording

- ECG: recording of the heart's electrical activity
- It is recorded through electrodes placed in the torso and limbs
- Electrodes measures the electrical activity of the heart from different orientations (leads)



Chest leads (unipolar)

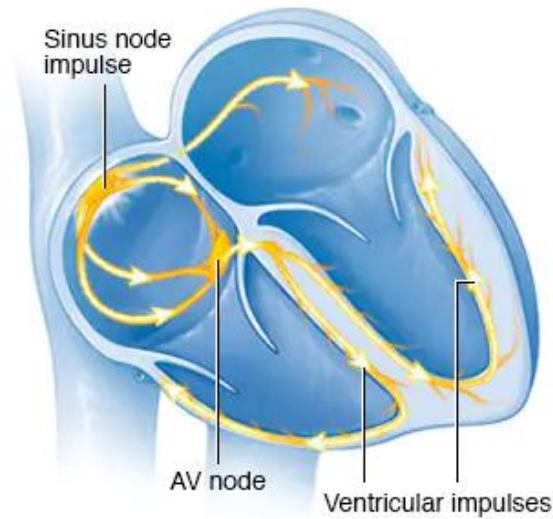


$$V1 = V1_{\text{elettrodo}} - \frac{RA + LA + LL}{3}$$

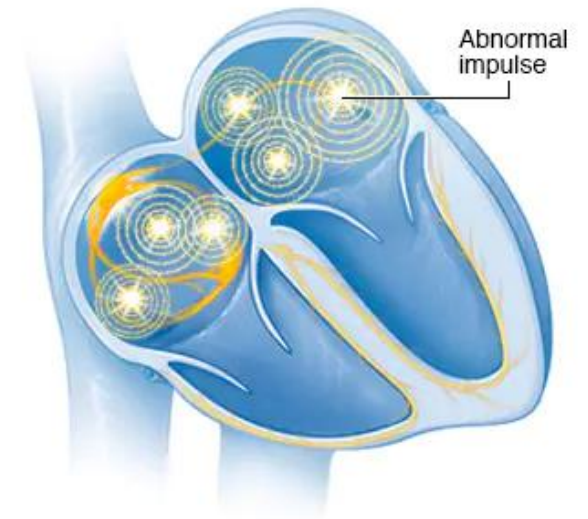
What is atrial fibrillation?

- Is the most common sustained arrhythmia, affecting around 2% of global population
- It increases the risk of **thrombus** formation and, consequently, of **stroke** and systemic **embolism**.
- Many times asymptomatic: diagnosed after an acute event.
- AI-based AF detection enables automatic and continuous ECG analysis

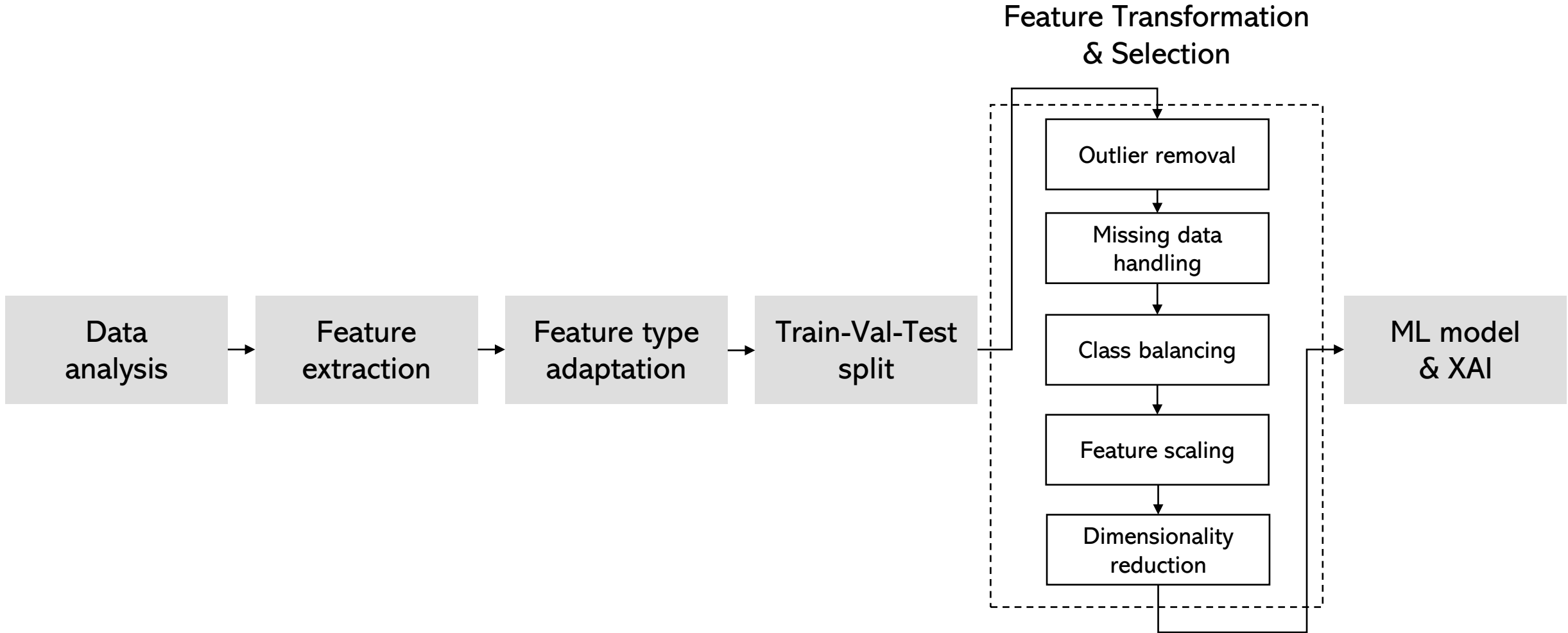
NSR: Normal sinus rhythm



AF: Atrial fibrillation



Workflow

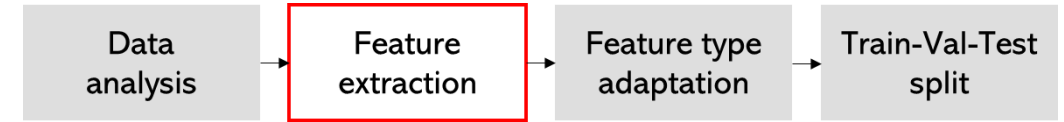


Data analysis

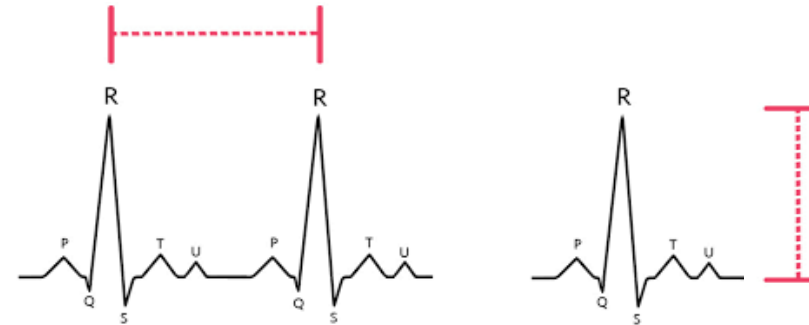
- **Label distribution**
 - Is the dataset balanced or unbalanced?
- **Signal length/duration**
 - What type of features can I extract?
- **Signal visualization**
 - How is the signal quality?
 - Are there any patterns?



Feature extraction



- **Patient features**
age, sex, sampling frequency...
- **Rhythm features**
RR duration, R peak amplitude
- **Morphological features**
vectorial analysis of the signal (provided)



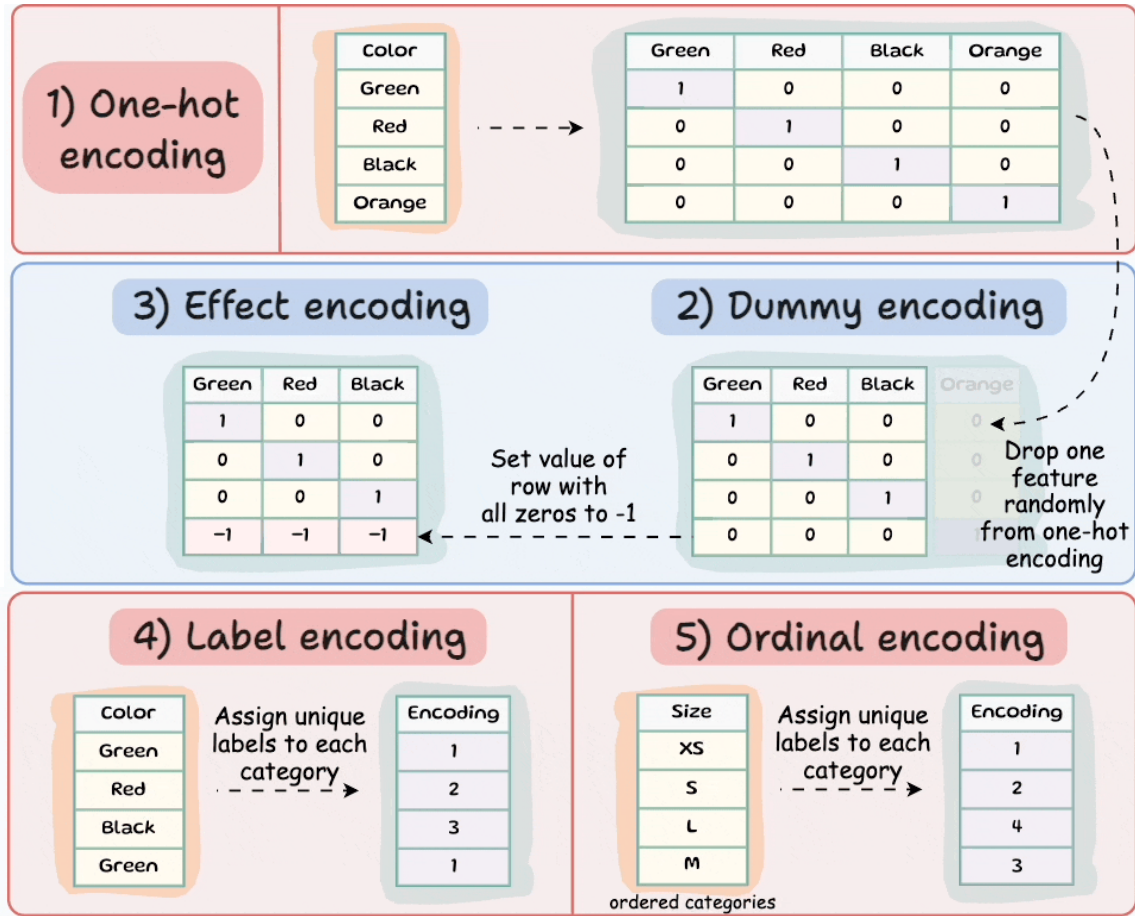
1. QRSTang:	peak mean QRS-T angle	
2. QRSTang_M:	QRS-T angle	
3. AZ_OQ:	Azimuth of QRS: Peak	
4. AZ_OQM:	Azimuth of QRS: area	
5. AZ_OT:	Azimuth of T:peak	
6. AZ_OTM:	Azimuth of T:area	
7. AZ_SVG:	Azimuth of SVG: Peak	
8. AZ_SVG_M:	Azimuth of SVG: area	
9. EL_OQ:	Elevation of QRS: Peak	
10. EL_OQM:	Elevation of QRS: area	
11. EL_OT:	Elevation of T: Peak	

12. EL_OTM:	Elevation of T: area	
13. EL_SVG:	Elevation of SVG:Peak	
14. EL_SVG_M:	Elevation of SVG: area	
15. QRS_Mag:	Magnitude of QRS: peak	
16. QRS_Mag_M	Magnitude of QRS: area	
17. T_Mag	Magnitude of T: peak	
18. T_Mag_M	Magnitude of T: area	
19. SVG_Mag	Magnitude of SVG: peak	
20. QT_interval		
21. AUC_VM_QT	AUC on Vector Magnitude	
22. WVG	Magnitude of WVG: Wilson's Ventricular Gradient	

Feature adaptation



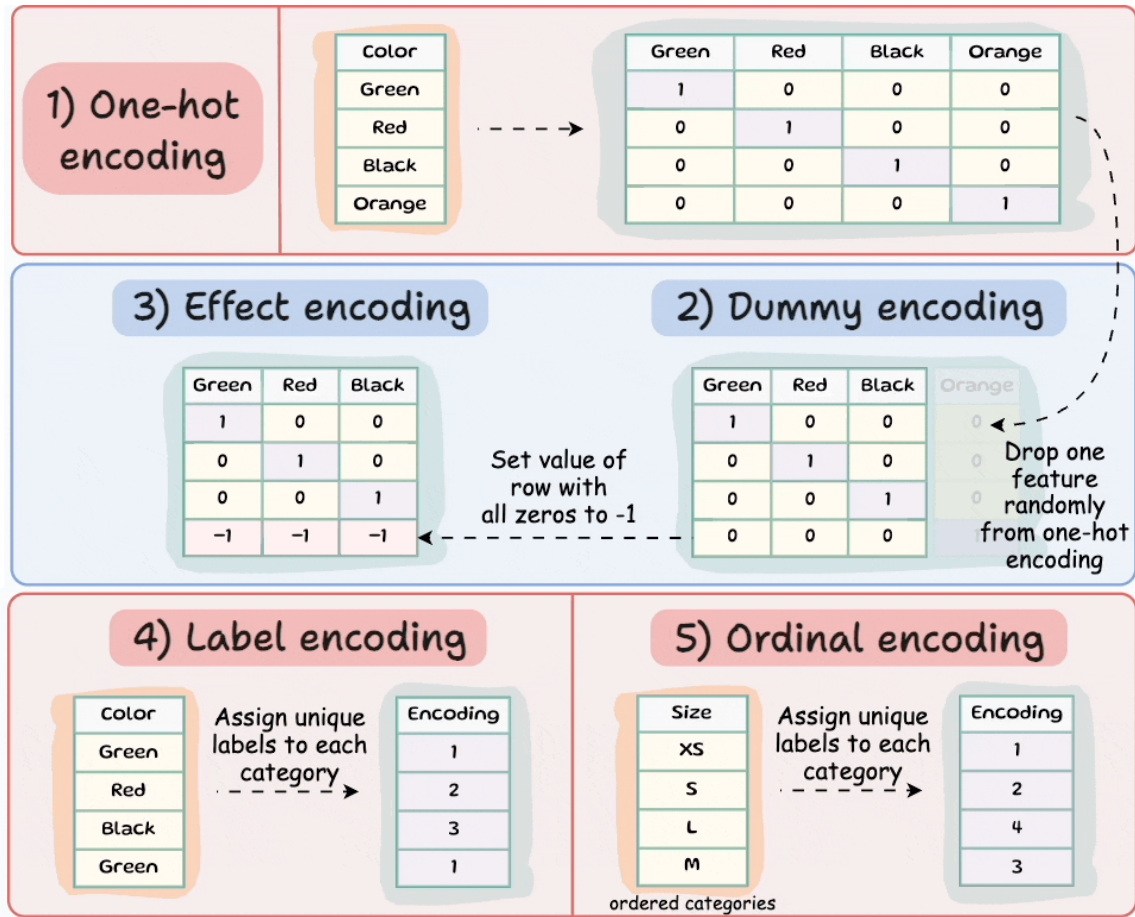
Categorical features should be adapted



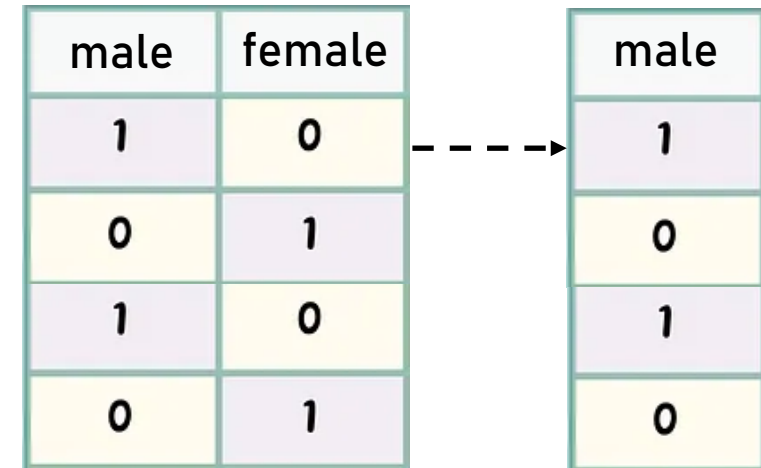
Feature adaptation



Categorical features should be adapted



BINARY CATEGORICAL FEATURES
e.g. sex → male/female



Data

760 subjects each with:

- header file .hea: subject information
- Matlab file .mat: 12-lead signals



leads Fs samples/lead → 10 sec

```
A0002 12 500 5000 12-May-2020 12:33:59
A0002.mat 16+24 1000/mV 16 0 -1 -1 0 I
A0002.mat 16+24 1000/mV 16 0 -15 22 0 II
A0002.mat 16+24 1000/mV 16 0 -15 -8 0 III
A0002.mat 16+24 1000/mV 16 0 7 5 0 aVR
A0002.mat 16+24 1000/mV 16 0 8 23 0 aVL
A0002.mat 16+24 1000/mV 16 0 -16 -21 0 aVF
A0002.mat 16+24 1000/mV 16 0 44 16 0 V1
A0002.mat 16+24 1000/mV 16 0 71 -29 0 V2
A0002.mat 16+24 1000/mV 16 0 30 -38 0 V3
A0002.mat 16+24 1000/mV 16 0 0 11 0 V4
A0002.mat 16+24 1000/mV 16 0 -5 49 0 V5
A0002.mat 16+24 1000/mV 16 0 -11 32 0 V6
#Age: 49
#Sex: Female
#Dx: 426783006
#Rx: Unknown
#Hx: Unknown
#Sx: Unknown
```

Data

1. Open this link

https://drive.google.com/drive/folders/1iIWQDOvwTz_kdCEx71_KfZhzxfbiOztO?usp=drive_link

2. Right click on the name of the folder → organize

3. Select add shortcut to the folder My Drive

4. In your Drive you should have this:

